

SIDDHANT RAI

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🐙 GitHub

🌐 LinkedIn

🌐 Portfolio

SUMMARY

Aspiring Machine Learning Engineer with practical experience building intelligent systems and deploying real-world applications. Created **Flashkick**, a football highlight generator using 3D CNNs, Whisper, and MoviePy. Built a **Movie Recommender System** using content-based filtering and deployed with Streamlit. Currently working on a **Book Recommender System** with collaborative filtering. Passionate about applying ML to solve real-world problems with clean, scalable code.

EDUCATION

Lalitpur Engineering College

Bachelor in Computer Engineering

2021–2025

Lalitpur, Nepal

SKILLS

Languages: Python, JavaScript, C++

Machine Learning Libraries: scikit-learn, pandas, NumPy, Seaborn, Matplotlib

Deep Learning: TensorFlow, Keras, CNNs, LSTM, GRU, RNNs, Embedding Layers, Sequence Modeling

Natural Language Processing: OpenAI Whisper, NLTK, CountVectorizer, TF-IDF

Recommender Systems: Content-Based Filtering, Collaborative Filtering, Matrix Factorization, Cosine Similarity

Web Frameworks: Flask, Streamlit

Tools and Platforms: Git, MoviePy, Google Colab, Heroku, Netlify, Kaggle

EXPERIENCE

Database Intern

Aqore Nepal

Jan 2025 – Mar 2025

Kathmandu, Nepal

PROJECTS

Kidney Disease Classification (DL) — TensorFlow, Keras, ResNet50, MLflow, DVC

Completed

- Built end-to-end DL pipeline classifying kidney CT scans into 4 classes with 84% validation accuracy.
- ResNet50 transfer learning with Softmax, Categorical Cross-Entropy, Adam optimizer (LR=0.0001).
- Employed MLOps: DVC pipelines, MLflow experiment tracking and model registry (v3).
- Developed prediction UI showing image name, predicted class, and confidence.

Flashkick Football Highlight Generator — React.js, Flask, 3D CNN

Completed

- 3D CNN model detecting key football moments with 92% accuracy.
- Optimized frame sampling to reduce highlight generation time by 45%.
- Deployed interactive web interface for real-time highlights.

Movie Recommender System — Content-Based Filtering, Streamlit

Completed

- Content-based movie recommender using MovieLens (100k ratings, 1,682 movies).
- Utilized CountVectorizer + cosine similarity for personalized recommendations.
- Deployed interactive Streamlit app for real-time evaluation.

AI Book Recommendation System — Semantic Search, LLM Embeddings, ChromaDB

Completed

- LLM-powered semantic search with MiniLM embeddings for intent-aware book recommendations.
- Emotion-aware ranking and category filtering for personalized results.
- Persistent vector database (ChromaDB) with HNSW similarity search and pagination.
- Streamlit UI with book details, lazy-loaded images, and similar book suggestions.

Next-Word Prediction using LSTM — TensorFlow, Keras, Quotes Dataset

Completed

- LSTM model predicting next word with 35% top-1 and 60% top-3 accuracy on Kaggle Quotes dataset (20k quotes).
- Implemented tokenization, padding, and sequence generation for training.
- Designed embedding + 2 LSTM layers with dense Softmax output.
- Deployed web demo for real-time prediction.

Vendor Purchase and Sales Data Analysis — SQL, Python

Completed

- Analyzed vendor performance with SQL & Pandas, highlighting trends and key metrics.
- Generated summaries and visualizations for business insights.
- Identified procurement optimization opportunities, projecting 15% cost reduction.

WORKSHOPS & TRAININGS

90-day Free Course – AI/ML (2025)

- Enforced end-to-end ML pipelines on small datasets, improving workflow efficiency by 20%.
- Applied predictive and recommender systems using realistic datasets, achieving 75% recommendation accuracy.
- Refined model evaluation and tuning processes, increasing performance on test sets by 10%.
- Deployed interactive ML applications using Streamlit, enabling 15 user interactions per session during testing.

45-day Free Course – Web Development (2025)

- Built 4 projects using JavaScript and React, increasing code reusability and reducing bugs by 15%.
- Used state management with React hooks and custom hooks to modularize components.
- Refined DOM manipulation, event handling, and CRUD operations to streamline UI updates.
- Executed data persistence with localStorage, reducing session reload issues by 10%.

MERN Training – Lalitpur Engineering College (2022)

- Built full-stack apps with MongoDB, Express, React, and Node.js, completing 1 project.
- Developed REST APIs and integrated frontend-backend for 3 modules, evaluated with 3 users.
- Gained experience with Git, improving version control practices during team exercises.

AWARDS & CERTIFICATIONS

- Certified AWS Cloud Foundation
- React.js Certificate — Complete Coding
- JavaScript Certificate — Complete Coding
- Certified AWS Cloud Practitioner from BroadwayInfosys

PARTICIPATION

- Poster Presentation at “International Conference on Engineering Technology & Infrastructure Development (2024)”